

COMMON EVOLUTIONARY MECHANISM OF ACTIVATION OF CLASS A GPCRS

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ABSTRACT

Cells communicate with their environment by integrating signals, often chemical in nature, triggered by specific molecules binding to specific membrane-bound receptors, resulting in a downstream signaling cascade. Arguably, G-protein-coupled receptors (GPCRs) constitute the most pharmacologically important family of such receptors, binding small molecules, peptides, lipids, and hormones with high specificity. However, despite a highly conserved fold and sequence similarity, GPCRs are still mostly studied on a case-by-case basis. Here, we infer a general, evolutionarily conserved mechanism of class A GPCR activation. By leveraging coevolution and machine learning methods applied to all class A GPCRs structures, we derive a mathematical description (a so-called collective variable - CV) of the receptor's activation state which is independent of its sequence. Then, we bias molecular dynamics simulations along this CV to obtain transitions between activation states of a diverse set of class A GPCR family members. To demonstrate that our model generalizes beyond GPCRs in our training set, we obtain conformational transitions of an orphan receptor, GPR183. Finally, we show that we can model ligand effect on the receptors by converging Free Energy Surfaces of activation of the β 2-adrenergic receptor within this common mechanism framework. These results, to our knowledge, prove for the first time the existence of a mechanism uniting all class A GPCRs. Our approach thus facilitates direct comparisons between receptors and opens up the possibility of structural and dynamical studies of many orphan and understudied GPCRs. It also serves as a blueprint for inferring family-wide protein mechanisms.